EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction to Polymers; Definition of Polymers; Classifications of Polymers; Source of Polymers; Structure of Polymers; Basis of Monomeric Species

1. Starch is a polymer of
(A) α-D-Glucose (B) β-D-Glucose
(C) α-D-Glucose and β-D-Glucose (D) α-D-Fructose

CPL091

2. Polymer which has amide linkage is
(A) Nylon-6, 6 (B) Terylene (C) Teflon (D) Bakelite

CPL092

3. Number average molecular mass $\left(\overline{M}_n\right)$ and weight average molar mass $\left(\overline{M}_w\right)$ of synthetic polymers are related as
(A) $\overline{M}_n < \overline{M}_w$ (B) $\overline{M}_n > \overline{M}_w$ (C) $\overline{M}_n = \overline{M}_w$ (D) $M_n > M_w$

CPL093

Classification of Polymers; Molecular Forces; Degradations; Mode of Polymerisation

4. Which of the following is wrong ?
(A) PMMA is called plexiglass (B) PTFE is called teflon
(C) SBR is called natural rubber (D) LDPE is called low density polyethylene

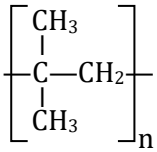
CPL094

Rakhi's Academy JEE

Examples of Addition Polymerisation; Difference Between LDPE and HDPE; Examples of Condensation Polymerisation

5. Ziegler-Natta catalyst is
(A) $K[PtCl_3(C_2H_4)]$ (B) $(Ph_3P)_3RhCl$ (C) $Al_2(C_2H_5)_6 + TiCl_4$ (D) $Fe(C_5H_5)_2$

CPL095

6. Monomer of given polymer  is

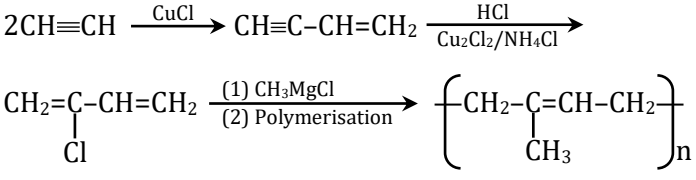
(A) 2-Methylpropene (B) Styrene (C) Propylene (D) Ethene

CPL096

7. Buna-S is a polymer of
(A) Butadiene only (B) Butadiene and nitrile
(C) Styrene only (D) Butadiene and styrene

CPL097

8. The polymerization reaction shown below



would produce
(A) PVC (B) neoprene (C) chloroprene (D) Rubber

CPL098

For More Join: @IITJEE_Advanced

9. Which of the following polymerises the most easily ?

(A) $CH_3CH_2C \equiv CH$ (B) $CH_2=CH-CH=CH_2$

(C) $CH_3CH_2-CH=CH_2$ (D) $CH \equiv C-C \equiv CH$

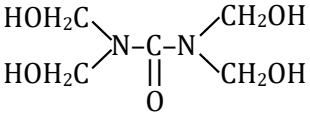
CPL099

10. Condensation product of caprolactam is

(A) nylon-6 (B) nylon-6, 6 (C) nylon-6, 0 (D) nylon-6, 10

CPL100

11.



The polymer obtained by the above compound is

(A) Bakelite (B) Urea formaldehyde resin

(C) Melamine formaldehyde resin (D) Teflon

CPL102

12. Which of the following is a condensation polymer ?

(A) Polystyrene (B) PVC (C) Polyester (D) Teflon

CPL103

13. Which of the following contains isoprene unit ?

(A) Natural-rubber (B) Polyethylene (C) Nylon-6, 6 (D) Dacron

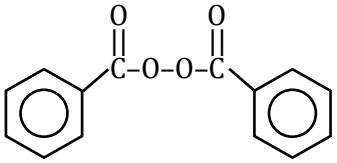
CPL105

Rankers Academy JEE

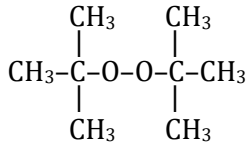
Mechanism of Free Radical Polymerisation Questions

14. Which of the following is radical initiator?

(A) $R - N = N - R$ (B)



(C)



(D) All

CPL101

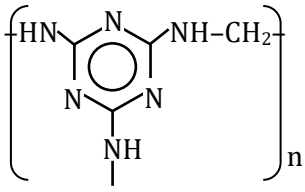
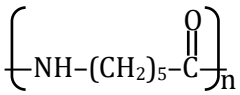
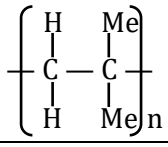
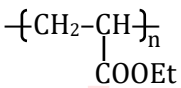
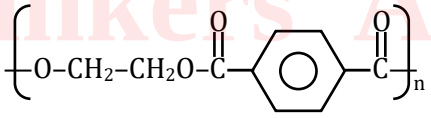
For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MATRIX MATCH TYPE QUESTION

Examples of Addition Polymerisation; Difference Between LDPE and HDPE; Examples of Condensation Polymerisation

1. Match the items given in column I with those in column II.

Column–I (Structure)		Column–II (Polymer)	
a.		p.	Polyisobutylene
b.		q.	PEA
c.		r.	Melmac
d.		s.	Dacron
e.		t.	Perlon

(A) a-r, b-t, c-p, d-q, e-s (B) a-r, b-t, c-p, d-s, e-q
(C) a-t, b-r, c-s, d-q, e-s (D) a-r, b-t, c-s, d-p, e-q

CPL107

Uses of Polymer

2. Match the items given in column I with those in column II.

Column I (Polymer)		Column II (Uses)	
a.	PHBV	p.	Non-sticky surface of cooking pans.
b.	PCTFE	q.	Transparent domes.
c.	PMMA	r.	Paints, lacquers
d.	Glyptal	s.	Control drug release

(A) a-s, b-p, c-r, d-q (B) a-s, b-r, c-q, d-p (C) a-s, b-p, c-q, d-r (D) a-r, b-q, c-p, d-s

CPL106

COMPREHENSION TYPE QUESTIONS

Classification of Polymers; Molecular Forces; Degradations; Mode of Polymerisation

For questions 3-5

The utility of polymers in various fields is due to their mechanical properties like tensile strength, elasticity, toughness, etc. These properties mainly depend upon intermolecular forces like van der Waals forces and hydrogen bonding operating in polymer molecules. Polymers have been classified on this basis as e.g.

(i) Elastomers (ii) Fibres (iii) Thermoplastics (iv) Thermosetting

Rankers Academy JEE
For More Join: @IITJEE_Advanced

3. The molecular forces of attraction are the weakest in

(A) elastomers (B) fibres (C) thermoplastics (D) thermosetting polymers

CPL108

4. Which of the following have usually a linear structure?

(A) Thermoplastics (B) Thermosetting polymers

(C) Bakelite (D) Melamine

CPL109

5. Which of the following is hardest?

(A) Elastomer (B) Fibre (C) Thermoplastic (D) Thermosetting polymers

CPL110

Mechanism of Free Radical Polymerisation Questions

For questions 6-8

Polymerisation involves various mechanisms depending upon the initiator. If initiator generates free radical, addition polymerisation takes place through free radical mechanism. If H^+ is initiator and electron releasing group is present, cationic polymerisation takes place. If electron withdrawing group is present and KNH_2 is initiator, anionic polymerisation takes place.

6. Which of the following will undergo anionic polymerisation in presence of KNH_2?

(A) $CH_2=\underset{\displaystyle C_6H_5}{\underset{|}{CH}}$ (B) $CH_2=\underset{\displaystyle CH_3}{\underset{|}{CH}}$ (C) $CH_2=\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{CH}}}}$ (D) $CH_2=CH_2$

Rankers Academy JEE

CPL111

7. Which of the following will induce anionic polymerisation?

(A) KNH_2 (B) n-Butyl Lithium (C) KOH (D) All of these

CPL112

8. Which of the following will undergo cationic polymerisation in presence of $H^{\oplus}$?

(A) $CH_2=\underset{\displaystyle Cl}{\underset{|}{CH}}$ (B) $CH_2=\underset{\displaystyle CN}{\underset{|}{CH}}$ (C) $CH_2=\underset{\displaystyle NO_2}{\underset{|}{CH}}$ (D) $CH_2=\underset{\displaystyle CH_3}{\underset{|}{CH}}$

CPL113

MULTIPLE CORRECT TYPE QUESTIONS

Examples of Addition Polymerisation; Difference Between LDPE and HDPE; Examples of Condensation Polymerisation

9. Structure of some common polymers are given. Which is/are correctly matched with its/their name ?

(A) Teflon $\left(-CF_2-CF_2-\right)_n$

(B) Neoprene $\left(-CH_2-\underset{\displaystyle Cl}{\underset{|}{C}}=CH-CH_2-CH_2-\right)_n$

(C) Terylene $\left(-OC-\bigcirc-COOCH_2-CH_2-O-\right)_n$

(D) Nylon-6,6 $\left(-NH(CH_2)_6NHCO(CH_2)_4-CO-\right)_n$

CPL114

For More Join: @IITJEE_Advanced

10. The correct statement(s) about peptides is/are

(A) A dipeptide has one peptide link between two amino acids.

(B) By convention N-Terminus is kept at left and C-terminus at right in the structure of a peptide

(C) If only one amino group and one carboxylic acid, group are available for reaction, then only one dipeptide can forms.

(D) A polypeptide with more than hundred amino acid residues (mol. mass > 10,000) is called a protein.

CPL115

11. Preparation of nylon from hexamethylene diamine and adipic acid is an example of

(A) addition polymerisation (B) homopolymerisation

(C) condensation polymerisation (D) copolymerisation

CPL116

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).
- Answer to each question will be evaluated according to the following marking scheme:
- Full Marks : +4, If correct answer is entered.
- Zero Marks : 0, If the question is unanswered.
- Negative Marks : –1, If wrong answer is entered.

Classification of Polymers; Molecular Forces; Degradations; Mode of Polymerisation

1. PDl for natural polymers is generally close to .....

CPL117

Mechanism of Free Radical Polymerisation Questions

2. How many of the following is/are a condensation polymer ?

(a) Polystyrene (b) Neoprene

(c) PAN (d) Polyethylene terephthalate

CPL119

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/ rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3, if ONLY the correct numerical value is entered as answer.

Zero Marks : 0, in all other cases.

Radius Academy JEE

Classification of Polymers; Molecular Forces; Degradations; Mode of Polymerisation

3. How many carbon atoms are present in a monomer of neoprene?

CPL029

Examples of Addition Polymerisation; Difference Between LDPE and HDPE; Examples of Condensation Polymerisation

4. Number of nitrogen atoms in a monomer unit of melamine is ?

CPL026

5. What will be the mass percentage of acrylonitrile in copolymer formed with vinylacetate (it contains alternate monomers of each type)?

CPL027

6. Poly β-hydroxybutyrate-co-β-hydroxyvalerate (PHBV) is obtained by the copolymerisation of two hydroxy acids (A) & (B)

(A) + (B) $\longrightarrow$ PHBV

Find the sum of number of chiral centres present in A & B

CPL028

7. How many sp^3 carbon atoms are present in caprolactam ?

CPL030

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Match List-I and List-II. **[JEE (Main) 2023]**

List-I	List-II
A. Elastomeric polymer	I. Urea formaldehyde resin
B. Fibre polymer	II. Polystyrene
C. Thermosetting polymer	III. Polyester
D. Thermoplastic polymer	IV. Neoprene

Choose the correct answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-III, C-I, D-II
(4) A-IV, B-I, C-III, D-II

CPL066

2. Match List I with List II **[JEE (Main) 2023]**

List-I	**List-II**
Polymer	**Type/Class**
(A) Nylon-2-Nylon-6	(I) Thermosetting Polymer
(B) Buna-N	(II) Biodegradable polymer
(C) Urea-formaldehyde	(III) Synthetic rubber resin
(D) Dacron	(IV) Polyester

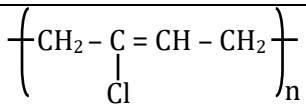
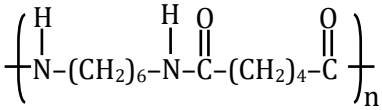
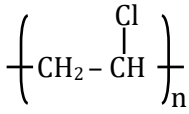
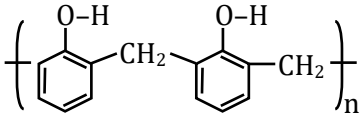
Choose the correct answer from the options given below:

(1) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)
(2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
(3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(4) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Learners Academy JEE

CPL127

3. Match **List I** with **List II.** **[JEE (Main) 2022]**

	List-I (Polymers)		**List-II (Nature)**
A.		I.	Thermosetting polymer
B.		II.	Fibers
C.		III.	Elastomer
D.		IV.	Thermoplastic polymer

Choose the correct answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-III, C-IV, D-II

CPL122

For More Join: @IITJEE_Advanced

4. Which of the following is **NOT** a natural polymer? **[JEE (Main) 2022]**

(1) Protein (2) Starch (3) Rubber (4) Rayon

CPL123

5. Caprolactam when heated at high temperature in presence of water, gives **[JEE (Main) 2023]**

(1) Teflon (2) Dacron (3) Nylon 6, 6 (4) Nylon 6

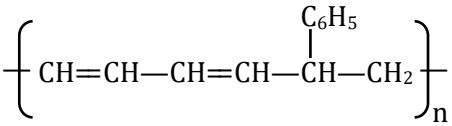
CPL067

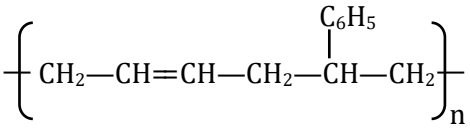
6. Polymer used in orlon is: **[JEE (Main) 2023]**

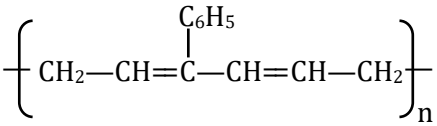
(1) Polyacrylonitrile (2) Polyethene (3) Polycarbonate (4) Polyamide

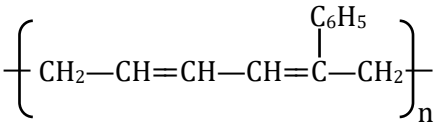
CPL126

7. Buna–S can be represented as: **[JEE (Main) 2023]**

(1) 

(2) 

(3) 

(4) 

CPL128

8. The polymer X – consists of linear molecules and is closely packed. It is prepared in the presence of triethylaluminium and titanium tetrachloride under low pressure. The polymer X is – **[JEE (Main) 2023]**

(1) Polyacrylonitrile (2) Low density polythene

(3) Polytetrafluoroethane (4) High density polythene

Rankers Academy JEE

CPL129

9. Given below are two statements : **[JEE (Main) 2023]**

Statements I : Ethene at 333 to 343K and 6-7 atm pressure in the presence of $AlEt_3$ and $TiCl_4$ undergoes addition polymerization to give LDP.

Statement II : Caprolactam at 533-543K in H_2O through step growth polymerizes to give Nylon 6.

In the light of the above Statements, chose the correct answer from the options given below :

(1) Both Statement I and Statement II are true.

(2) Statement I is false but Statement II is true.

(3) Statement I is true but Statement II is false.

(4) Both Statement I and Statement II are false.

CPL130

10. Match the following **[JEE (Main) 2023]**

Column –A		**Column-B**	
a	Nylon 6	**I**	Natural Rubber
b	Vulcanized Rubber	**II**	Cross Linked
c	cis-1,4-polyisoprene	**III**	Caprolactam
d	Polychloroprene	**IV**	Neoprene

Choose the correct answer from option given below:

(1) $a \rightarrow IV, b\text{-}III, c \rightarrow II, d \rightarrow I$ (2) $a \rightarrow III, b \rightarrow IV, c \rightarrow I, d \rightarrow II$

(3) $a \rightarrow II, b \rightarrow III, c \rightarrow IV, d \rightarrow I$ (4) $a \rightarrow III, b \rightarrow II, c \rightarrow I, d \rightarrow IV$

CPL131

For More Join: @IITJEE_Advanced

11. Match List I with List II **[JEE (Main) 2023]**

LIST–I		LIST–II	
A.	Weak intermolecular forces of attraction	I.	Hexamethylenediamine + adipic acid
B.	Hydrogen bonding	II.	$AlEt_3 + TiCl_4$
C.	Heavily branched polymer	III.	2–chloro–1, 3-butadiene
D.	High density polymer	IV.	Phenol + formaldehyde

Choose the correct answer from the options given below:

(1) A-II, B-IV, C-I, D-III
(2) A-III, B-I, C-IV, D-II
(3) A-IV, B-I, C-III, D-II
(4) A-IV, B-II, C-III, D-I

CPL132

12. Match List I with List II: **[JEE (Main) 2023]**

List I-(Monomer)	**List II-(Polymer)**
(A) Tetrafluoroethene	(i) Orlon
(B) Acrylonitrile	(ii) Natural rubber
(C) Caprolactam	(iii) Teflon
(D) Isoprene	(IV) Nylon-6

Choose the correct answer from the options given below :

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
(2) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
(4) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

CPL133

13. Which of the following is **not** an example of a condensation polymer? **[JEE (Main) 2022]**

(1) Nylon 6,6
(2) Dacron
(3) Buna-N
(4) Silicone

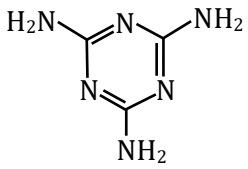
CPL062

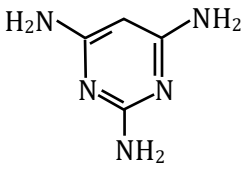
14. Which of the following is an example of polyester? **[JEE (Main) 2022]**

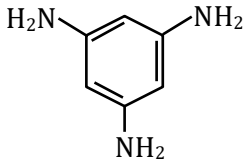
(1) Butadiene-styrene copolymer
(2) Melamine polymer
(3) Neoprene
(4) Poly-β-hydroxybutyrate-co-β-hydroxy valerate

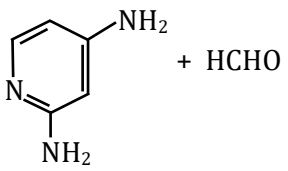
CPL063

15. Melamine polymer is formed by the condensation of : **[JEE (Main) 2022]**

(1)  + HCHO

(2)  + HCHO

(3)  + HCHO

(4)  + HCHO

CPL064

For More Join: @IITJEE_Advanced

16. Match **List-I** with **List-II**. **[JEE (Main) 2022]**

List-I (Polymer)	**List-II (Monomer)**
(A) Neoprene	(I) Acrylonitrile
(B) Teflon	(II) Chloroprene
(C) Acrilan	(III) Tetrafluoroethene
(D) Natural rubber	(IV) Isoprene

Choose the correct answer from the option given below:

(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV) (2) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

(3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III) (4) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

CPL120

17. Terylene polymer is obtained by condensation of : **[JEE (Main) 2022]**

(1) Ethane-1, 2-diol and Benzene-1, 3 dicarboxylic acid

(2) Propane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

(3) Ethane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

(4) Ethane-1, 2-diol and Benzene-1, 2 dicarboxylic acid

CPL121

18. Match List I with List II

	List-I (Name of polymer)		**List-II** (Uses)
A.	Glyptal	I.	Flexible pipes
B.	Neoprene	II.	Synthetic wool
C.	Acrilan	III.	Paints and Lacquers
D.	LDP	IV.	Gaskets

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Choose the correct answer from the options given below :- **[JEE (Main) 2023]**

(1) A-III, B-II, C-IV, D-I (2) A-III, B-IV, C-II, D-I

(3) A-III, B-IV, C-I, D-II (4) A-III, B-I, C-IV, D-II

CPL068

19. Match List I with List II : **[JEE (Main) 2022]**

	List-I Polymer		**List-II used for items**
A.	Nylon 6,6	I.	Buckets
B.	Low density polythene	II	Non-stick utensils
C.	High density polythene	III.	Bristles of brushes
D.	Teflon	IV.	Toys

Choose the correct answer from the options given below:

(1) A–III, B-I, C-IV, D-II (2) A–III, B-IV, C-I, D-II

(3) A–II, B-I, C-IV, D-III (4) A–II, B-IV, C-I, D-III

CPL065

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Among the following, the correct statement(s) about polymers is(are) **[JEE (Advanced) 2022]**

(A) The polymerization of chloroprene gives natural rubber.

(B) Teflon is prepared from tetrafluoroethene by heating it with persulphate catalyst at high pressures.

(C) PVC are thermoplastic polymers.

(D) Ethene at 350-570 K temperature and 1000-2000 atm pressure in the presence of a peroxide initiator yields high density polythene.

CPL135

2. Choose the correct option(s) from the following **[JEE (Advanced) 2019]**

(A) Natural rubber is polyisoprene containing *trans* alkene units

(B) Nylon-6 has amide linkages

(C) Cellulose has only α-D-glucose units that are joined by glycosidic linkages

(D) Teflon prepared by heating tetrafluoroethene in presence of a persulphate catalyst at high pressure

CPL134

3. On complete hydrogenation, natural rubber produces **[JEE (Advanced) 2016]**

(A) Ethylene-propylene copolymer (B) Vulcanised rubber

(C) Polypropylene (D) Polybutylene

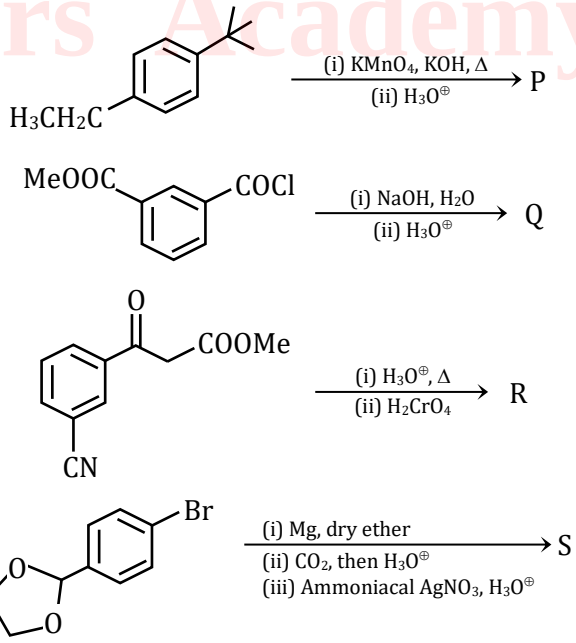
CPL090

4. The total number of lone-pairs of electrons in melamine is **[JEE (Advanced) 2013]**

CPL089

5. In the following reactions, **P, Q, R,** and **S** are the major products.

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The correct statement (s) about **P, Q, R,** and **S** is (are) **[JEE (Advanced) 2023]**

(A) **P** and **Q** are monomers of polymers Dacron and glyptal, respectively.

(B) **P, Q,** and **R** are dicarboxylic acids.

(C) Compounds **Q** and **R** are the same.

(D) **R** does **not** undergo aldol condensation and **S** does **not** undergo Cannizzaro reaction.

CPL136

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	A	C	C	A	D	D	B	A
Que.	11	12	13	14						
Ans.	B	C	A	D						

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	A	A	D	A	D	D	ACD	ABCD
Que.	11									
Ans.	CD									

EXERCISE (S)

Que.	1	2	3	4	5	6	7
Ans.	1	1	4.00	6.00	38.129	2.00	5.00

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	4	2	4	4	1	2	4	2	4
Que.	11	12	13	14	15	16	17	18	19	
Ans.	2	1	3	4	1	1	3	2	2	

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	
Ans.	BC	BD	A	6	CD	

For More Join: @IITJEE_Advanced

Important Notes

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For More Join: @IITJEE_Advanced